

PRACTICAL – 6

'''

Implement any one of the following Expert System

I. Information management

II. Hospitals and medical facilities

III. Help desks management

IV. Employee performance evaluation

V. Stock market trading

VI. Airline scheduling and cargo schedules

'''

Simple Help Desk Expert System

```
print("Welcome to Help Desk Expert System!")
```

```
print("Answer a few questions about your issue to get help.\n")
```

Collect user inputs

```
issue_type = input("\n Is your issue related to 'software', 'hardware', or 'network'? ").strip().lower()
```

```
urgency = input("\n Is this issue 'urgent' or 'normal'? ").strip().lower()
```

```
user_level = input("\n Are you a 'beginner' or 'advanced' user? ").strip().lower()
```

Rule-based decision

```
if issue_type == "software":
```

```
    if urgency == "urgent":
```

```
        solution = "Contact IT support immediately to resolve software issue."
```

```
    else:
```

```
        solution = "Try restarting the application or reinstalling the software."
```

```
elif issue_type == "hardware":
```

```
    if user_level == "beginner":
```

```
        solution = "Check cables, power supply, and basic hardware connections."
```

```
    else:
```

```
        solution = "Run hardware diagnostics or contact the hardware support team."
```

```
elif issue_type == "network":
```

```
    if urgency == "urgent":
```

```
        solution = "Contact network administrator immediately."
```

```
    else:
```

```
        solution = "Check your Wi-Fi connection and router settings."
```

```
else:
```

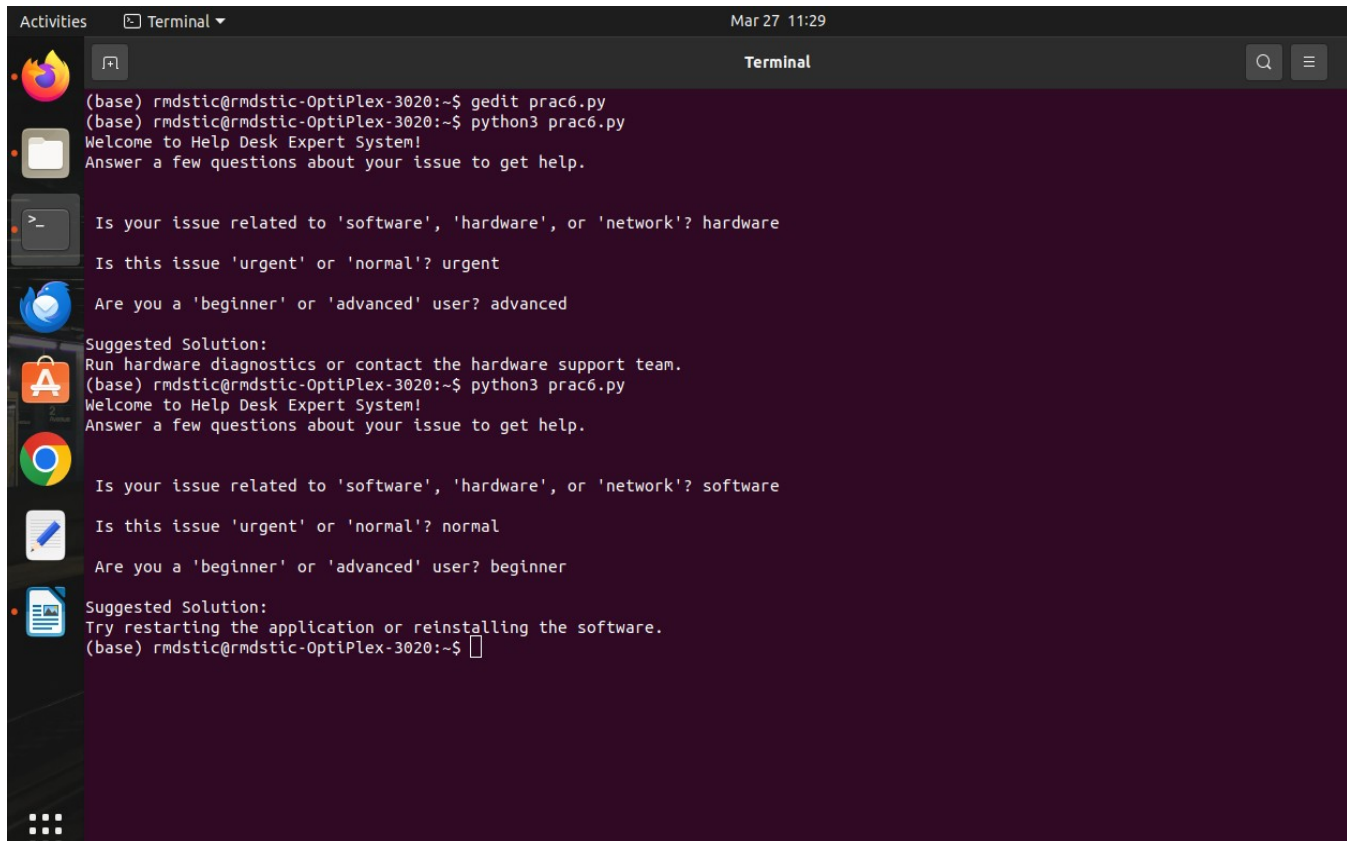
```
    solution = "Please contact the help desk for further assistance."
```

Display solution

```
print("\nSuggested Solution:")
```

```
print(solution)
```

OUTPUT



```
(base) rmdstic@rmdstic-OptiPlex-3020:~$ gedit prac6.py
(base) rmdstic@rmdstic-OptiPlex-3020:~$ python3 prac6.py
Welcome to Help Desk Expert System!
Answer a few questions about your issue to get help.

Is your issue related to 'software', 'hardware', or 'network'? hardware

Is this issue 'urgent' or 'normal'? urgent

Are you a 'beginner' or 'advanced' user? advanced

Suggested Solution:
Run hardware diagnostics or contact the hardware support team.
(base) rmdstic@rmdstic-OptiPlex-3020:~$ python3 prac6.py
Welcome to Help Desk Expert System!
Answer a few questions about your issue to get help.

Is your issue related to 'software', 'hardware', or 'network'? software

Is this issue 'urgent' or 'normal'? normal

Are you a 'beginner' or 'advanced' user? beginner

Suggested Solution:
Try restarting the application or reinstalling the software.
(base) rmdstic@rmdstic-OptiPlex-3020:~$
```